

PARVESH SHANMUGAM

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EDUCATION

B.Tech – CSE (Cyber Physical Systems), VIT Chennai | CGPA: 8.35

2023 – 2027

Class XII – Velammal Vidhyalaya Annexure, Chennai | **90.4%**

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript, TypeScript

Databases: MySQL, PostgreSQL, SQLite, MongoDB

Web & Frameworks: React, Next.js, Node.js, Express.js, Django, FastAPI, Spring Boot, MERN Stack

Tools: Git, GitHub, Linux, Postman, Docker, VS Code, IntelliJ

AI/ML & Generative AI : Machine Learning, Deep Learning, CNN, LSTM, GNN, YOLO, Federated Learning, XAI, LLMs, RAG

Security: Blockchain, Post-Quantum Cryptography, IoT Security, Edge AI, AI-based IDS

RESEARCH & PUBLICATIONS

Secure Smart City Data-Sharing Platform

Flask, REST APIs | Book Chapter 2025

- Privacy-preserving data exchange reducing integration time by 30% and latency by 25%; co-authored CPS book chapter on Critical Infrastructure Protection.

FedUnlearnGuard – Byzantine-Resilient Federated Unlearning

PyTorch, Docker, Hyperledger

- GDPR-compliant federated learning framework with 4-layer defense; reduced poisoning attacks from 92% to 3% with minimal accuracy loss.

Physics-Aware GPS Spoofing Detection for UAVs

ML, Explainable AI

- Interpretable trajectory-physics framework achieving ~95% spoofing detection via motion-constraint feature engineering and real-time streaming.

Automated OCT Skin Layer Segmentation for Diabetic Foot Analysis Imaging | Journal Paper Under Review

Deep Learning, U-Net Architectures, Biomedical

- Developed a deep learning pipeline using U-Net-based architectures for automated segmentation of OCT skin layers and quantitative analysis of microstructural changes associated with diabetic foot.

Fairness-Aware Quantum-Inspired Optimization for Smart City Digital Twins

QIGA, OpenDSS

- Multi-objective quantum-inspired framework for equitable infrastructure resource allocation across smart city systems.

Quantum-Resistant Blockchain Framework for IoT-Edge Systems

Post-Quantum Crypto, AI IDS

- Hardware-software co-design: lattice-based cryptography + blockchain authentication + AI IDS + vision-based monitoring for secure autonomous systems.

WORK EXPERIENCE

Research & Development Intern – VIT Chennai (Smart City)

May – Jul 2024 | Virtual

- Built Flask REST API platform cutting integration time by 30% and latency by 25%; co-authored Cyber Physical Systems academic book chapter.

Research Assistant – VIT Chennai (Medical Imaging & ML)

Mar 2025 – Present | Virtual

- Annotating OCT and thermal foot images; training ML models to detect diabetic foot ulcers and support clinical severity assessment.

Full-Stack Developer Intern – ennVee Solutions Pvt. Ltd. (<https://ennvee.com/>)

May – Jun 2025 | On-site

- Delivered Skills Matrix Portal (Django/MySQL) and role-based Task Management System (React/FastAPI/MySQL); analyzed Viva Learning adoption via Jira & Rally.

Full-Stack Developer Intern – IXES Global Solutions (<https://ixes.ai/>)

Sep – Nov 2025 | Hybrid

- Built production full-stack applications; handled database design, REST APIs, and dynamic UI with focus on scalability and performance.

SIEM & Security Operations Intern – Thinksynq Solutions Pvt. Ltd. (<https://thinksynq.in/>)

May – Jun 2026 | On-site

- Administered and analyzed the Thinklog SIEM platform by managing logs, alerts, and security monitoring, while developing custom queries, integrating systems through log forwarding, and monitoring security events across enterprise network and server infrastructure.

PROJECTS

MR GLOSS (Next.js, Three.js, SEO) – High-performance business site with 3D animations, chatbot integration, and SEO optimization (<https://www.mrglossceramiccoating.com/>).

MERCARA EXPORTS (React, Tailwind, Full Stack) – Scalable B2B export platform with product catalogs and premium UI for agro-commodity operations.

Dental Clinic Management – Dr. Bharads (Full Stack) – Production app for appointment handling, service management, and clinic digital presence (<https://www.smileywood.in/>).

ZYPH26 Summit App (Full Stack) – Official event application powering the Nilgiris Calls Global Summit at KCT Tech Park, Coimbatore (<https://zyph26.texus.io/>).

AI-Driven Nursing Competency Evaluation and Adaptive Training System (LLMs) – AI-powered platform that assesses nurses with adaptive quizzes and automatically generates personalized learning modules to improve their clinical competencies.

ACHIEVEMENTS

- Co-authored book chapter: "**Critical Infrastructure Protection: CPS Stability, Security, Privacy & Forensics**" – publication 2025.
- Best Paper Award – "**AI-Powered Traffic Signals and Street Lights**" at International SCES 2024 (Feb 2024).
- Proud Sponsor, Nilgiris Calls Global Summit | ZYPH Hackathon, KCT Tech Park – built and delivered zyph26.texus.io.